

Humidity Indicator and Controller

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Presented here is a simple humidity indicator and controller. In industries such as textile, the change in moisture content has a direct impact on the properties of fabric, such as tensile strength, elasticity, fibre diameter and friction. Therefore the process is executed within a permissible humidity environment only. Depending on the type of fabric and the process being undertaken, the requirement of particular humidity level varies. Cotton and linen have to be processed at very high relative humidity (RH) levels of around 70-80 per cent, since they are very brittle. Wool requires RH levels of around 65 per cent. Silk needs to be processed between 65 and 70 per

cent. With this circuit, you can not only monitor humidity levels between 30 and 90 per cent RH but also control it.



Fig. 2: Humidity sensor module

Circuit and working

The circuit diagram of humidity indicator and controller is shown in Fig. 1. Humidity sensor SY-HS-220 is used for sensing humidity levels and LM3914 dot/bar display driver to drive the LEDs (LED1 through LED7) corresponding to sensed humidity levels. The LEDs indicate the humidity level from 30 per cent RH to 90 per cent RH, as shown in the circuit. Transistors T1 and T2 are relay drivers through which electrical devices can be switched on or off.

Humidity sensor (SY-HS-220). Humidity sensors are gaining popularity in the field of measurement and control technology. The sensors for measuring water vapours as RH can be classified by the

Test Points

Test point	Details
TP0	0V, GND
TP1	5V
TP2	Low when LED1 glows

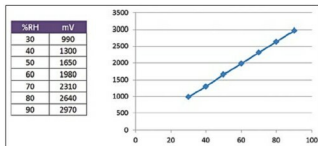


Fig. 3: Relative humidity (RH) versus output voltage

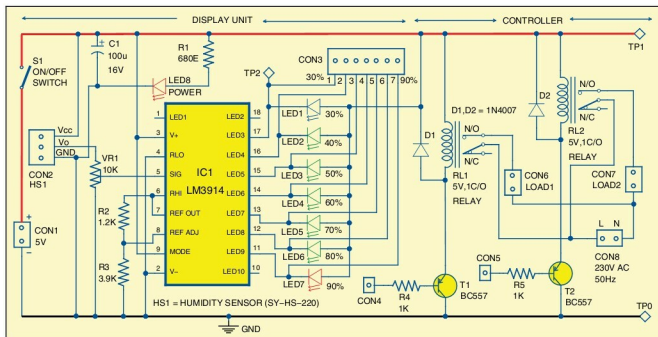


Fig. 1: Circuit diagram of humidity indicator and controller